

PROFESSIONAL EXPERIENCE

Automation Engineer | Sanofi Pasteur, Waltham, MA

Jun 2025 – Present

- Built a support ticketing platform using reactJS for 100+ staff, replacing ad-hoc verbal requests with tracked tickets that route to a team of engineers and trigger email alerts via SMTP.
- Designed a POC WebSocket-based dashboard that streams ticket status and resolution progress live, giving the team real-time visibility into open workload.
- Architected an ingestion pipeline that parses raw instrument metrics and logs into structured records on Prometheus and Loki, powering Grafana dashboards to track performance and monitor run state in Grafana dashboards
- Introduced Git version control and code review workflows to a previously unversioned production codebase, establishing release traceability and rollback.

Automation Engineer | Foundation Medicine Inc., Boston, MA

Nov 2024 – Apr 2025

- Designed and delivered ETL pipelines that ingest, validate, and transform sample records into a Sapio LIMS backend, automating tracking from intake to processing and eliminating a primary source of manual data-entry errors.
- Surfaced device configuration metadata through a React application that parses and renders it on demand, cutting troubleshooting time 40% by making opaque system parameters directly inspectable.
- Shipped a self-service Python desktop tool (CustomTkinter) exposing backend configuration management through a guided UI, cutting support requests by 20%.

Automation Engineer (contractor) | Sanofi Pasteur, Waltham, MA

Oct 2022 – Nov 2024

- Engineered a custom Python toolkit for automated parsing, transformation, and report generation across high-throughput instrument outputs, standardizing data formats across 4 teams.
- Integrated 10+ distributed hardware devices into orchestrated workflows over TCP/IP and REST APIs, diagnosing and resolving communication failures to keep data exchange reliable across the fleet
- Optimized core scheduling routines in a legacy codebase by batching redundant hardware operations that had been coded step-by-step, cutting end-to-end turnaround time 30% and improving output consistency.

Senior Research Associate | Dermabion Inc., Boston, MA

Oct 2021 – Sep 2022

- Operated production data pipelines processing high-throughput genomic datasets across Linux and HPC environments and created containerized QC pipelines alongside software engineers with automated data-quality checks that reduced processing errors 25%.

PROJECTS

E-Commerce Analytics Pipeline — Airflow 3, PySpark, dbt, GCS, BigQuery, Docker

- Engineered a DAG Airflow pipeline where a PySpark ingestion layer cleans and enriches 25,000+ records into Parquet on GCS, then triggers a dbt layer via ExternalTaskSensor to materialize revenue and customer KPIs in BigQuery.

Self-hosted Kubernetes Homelab & Observability Stack— Kubernetes, Prometheus, Grafana, Loki, Alloy, ArgoCD

- Provisioned a 3-node Kubernetes cluster with kubeadm, hand-configuring control plane, CNI networking, and persistent storage to run PostgreSQL, dedicated game servers, and network services alongside the observability stack.
- Instrumented cluster and workloads with Prometheus and Alloy, authoring custom Grafana dashboards and Alertmanager rules that page a Discord webhook on projected server's uptime, memory usage, and other user-visible symptoms.
- Implemented GitOps-based service management with ArgoCD, tracking and versioning all manifests in Git to enable continuous reconciliation and single-command rollback to any prior versions

EDUCATION

Northeastern University, Boston, MA — Graduate Certificate, Bioinformatics (Software & Data Mgmt)

May 2025

Uniwersytet Jagielloński, Krakow, Poland — M.S., Molecular Biotechnology

Jul 2021

University of California, Irvine, Irvine, CA — B.S., Pharmaceutical Sciences

Sep 2019

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Bash, R, HTML/CSS

Full Stack & Web: React, Node.js, Express, Electron, REST APIs, WebSockets, Tailwind CSS

Cloud & DevOps: Kubernetes, Docker, Terraform, GitOps (ArgoCD), AWS (S3), GCP (BigQuery, GCS), Git/GitHub, Linux

Data Engineering: Apache Airflow 3, PySpark, dbt, Kestra, BigQuery, PostgreSQL, MySQL, Parquet

Observability & BI: Prometheus, Grafana, Grafana Loki, Alloy, Alertmanager, Power BI, Looker Studio